

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

WINTER SEMESTER, 2024–2025
FULL MARKS: 75

CSE 4511: Computer Networks

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. Consider the network for "AllIutians," a startup based in Gazipur, which has been assigned the IPv4 address block 200.10.20.0/24. The network administrator must allocate IP addresses to the following departments based on their host requirements: Engineering (55 hosts), Sales (28 hosts), Management (10 hosts), and a WAN Link (2 IPs). **(29 Marks)**

- a) Design an optimal addressing scheme using Variable Length Subnet Masking (VLSM) to satisfy the requirements. Explicitly state the Subnet Mask, Network Address, and Broadcast Address for all the departments.

3 × 3
(CO4)
(PO3)

Solution:

Engineering (Need 64 IPs): 200.10.20.0/26 (Mask: 255.255.255.192, Bcast: 200.10.20.63)

Sales (Need 32 IPs): 200.10.20.64/27 (Mask: 255.255.255.224, Bcast: 200.10.20.95)

Management (Need 16 IPs): 200.10.20.96/28 (Mask: 255.255.255.240, Bcast: 200.10.20.111)

WAN (Need 4 IPs): 200.10.20.112/30 (Mask: 255.255.255.252, Bcast: 200.10.20.115)

Rubric:

1. For each Subnet Mask, Network Address, and Broadcast Address - 1 mark

- b) Calculate the total number of assignable IP addresses in the original /24 block versus the total number of IPs actually allocated to the subnets. Quantify the "wasted" IPs and justify if this remainder is sufficient for a future department requiring 20 hosts.

5
(CO1)
(PO1)

Solution:

Total IPs in /24 block: 256. Allocated: 64 + 32 + 16 + 4 = 116. Remaining: 256 – 116 = 140. A new department with 20 hosts requires a /27 block (32 IPs). Since 140 > 32, the remainder is sufficient to accommodate this future requirement.

Rubric:

1. Correct allocated ip addresses - 2.5 marks
2. Correct remaining ip addresses - 2.5 marks

- c) A new client connects to the Engineering subnet and requests an IP address. Briefly explain the sequence of DHCP messages used when a client joins the network. Which of these messages are broadcast and why?

5
(CO1)
(PO1)

Solution:

Sequence: Discover, Offer, Request, Acknowledge (DORA). The **Discover** and **Request** messages are broadcast because the client does not yet have an IP address and does not

know the specific IP address of the DHCP server to send a unicast. Request and Acknowledge can also be broadcast for the first time.

Rubric:

1. Explaining DORA - 2.5 marks
2. Reasoning behind broadcast DHCP - 2.5 marks

- d) The startup utilizes Network Address Translation (NAT) to interface with the Internet. Suppose an internal host (192.168.1.5:4000) sends a packet to a web server (8.8.8.8:80), explain how the router modifies the header (Source IP/Port) in the outgoing packet and how it knows where to send the incoming reply.

5
(CO1)
(PO1)

Solution:

Outgoing: The router replaces the private Source IP (192.168.1.5) with its public WAN IP and the Source Port (4000) with a unique ephemeral port. **Incoming:** The router checks its NAT translation table for the destination port of the reply packet to map it back to the original internal IP and port (192.168.1.5:4000).

- e) If the "AllIutians" network scales to 50 routers, which protocol between OSPF and RIP would offer faster convergence and lower bandwidth overhead? Justify your choice.

5
(CO2)
(PO2)

Solution:

OSPF is the better choice. RIP has a hop count limit of 15 (unsuitable for 50 routers) and slow convergence. OSPF uses Link State logic and hierarchical approach for fast convergence and only sends updates on changes, saving bandwidth compared to RIP's periodic broadcasts.

Rubric:

1. Why OSPF - 2.5 marks
2. Why not RIP - 2.5 marks

2. Suppose IUT CSE laboratory network relies on Layer 2 Ethernet switches to interconnect various departmental workstations. **(18 Marks)**

- a) A switch receives an Ethernet frame with Source MAC AA:AA and Destination MAC BB:BB. Assume the switch's forwarding table is currently empty. Explain the steps the switch takes to handle this frame. Explain the reason of this table having a timeout (aging time).

5 + 3
(CO1)
(PO1)

Solution:

1. **Learning:** Switch records Source MAC AA:AA and the ingress port in its table. 2. **Flooding:** Since Dest MAC BB:BB is unknown, it floods the frame out all other ports. Timeout exists to remove stale entries (e.g., if a device moves or disconnects), ensuring the table stays current and memory is not wasted.

Rubric:

1. Explaining Flooding - 5 marks
2. Timeout reasoning - 3 marks

- b) Student A wants to send data to Student B on the same LAN but only knows B's IP address. Explain how ARP resolves this.

5
(CO1)
(PO1)

Solution:

Student A broadcasts an ARP Request ("Who has IP B?"). Student B receives it and sends a unicast ARP Reply ("I am IP B, my MAC is X"). A then caches B's MAC address to send frames directly.

- c) You configure the switch to separate traffic into VLAN 10 (Students) and VLAN 20 (Faculty). Explain the concept of a Broadcast Domain. How does creating these two VLANs affect the number of broadcast domains?

5
(CO2)
(PO2)

Solution:

A Broadcast Domain is the set of devices that receive a broadcast frame originating from any device within the set. By default, a switch is 1 broadcast domain. Creating 2 VLANs splits the network into 2 separate broadcast domains (one for Student, one for Faculty).

Rubric:

1. Broadcast domain explaining - 2.5 marks
2. Effect of VLANS in broadcast domain in the given scenario - 2.5 marks

3. a) An IPv4 datagram with a Total Length of 4000 bytes (including a standard 20-byte IP header) arrives at a router. The datagram must be forwarded onto a link with an MTU (Maximum Transmission Unit) of 1500 bytes. The router must fragment this packet.
- i. Calculate and list the values for the following fields for each of the resulting fragments: Total Length, More Fragments (MF) Flag (0 or 1), Fragment Offset

9 + 5
(CO3)
(PO2)

Solution:

Max Data/Frag = 1480 bytes. Total Data = 3980 bytes.

Frag 1: Total Length 1500, MF=1, Offset=0.

Frag 2: Total Length 1500, MF=1, Offset=185 (1480/8).

Frag 3: Total Length 1040 (1020 data + 20 head), MF=0, Offset=370 (2960/8).

Rubric:

1. For each correct value - 1 mark

- ii. Why is the Identification (16-bit) field critical in this process? What would happen at the destination if two different source hosts accidentally used the same Identification number for their packets simultaneously?

Solution:

Identification links fragments to their original datagram for reassembly. If two *different* source hosts use the same ID, nothing bad happens because reassembly relies on the combination of Source IP, Dest IP, Protocol, ID. Since the Source IPs differ, the destination treats them as separate flows.

Rubric:

1. Reasoning behind identification field - 2 marks
2. Answer of the scenario - 3 marks

- b) A router receives a packet with TTL = 1. Explain exactly what the router would do now?

6
(CO1)
(PO1)

Solution:

The router decrements the TTL to 0. Since TTL is zero, it discards the packet to prevent routing loops. It then generates and sends an ICMPv4 "Time Exceeded" error message back to the source IP address.

Rubric:

1. Decrementing and dropping - 3 marks
2. Generating ICMP - 3 marks

- c) Compared to IPv4, several header fields have been removed and several new fields have been introduced in IPv6. Identify the major IPv4 header fields that were dropped in IPv6 and explain why they were dropped and how their functionalities are handled in the IPv6 protocol. Additionally, discuss the newly added IPv6 header fields and justify the purpose of each with appropriate reasoning.
- 8
(CO1)
(PO1)

MARKS SUMMARY TABLE

Question	Marks	Keywords
1.a	9	IPv4 addressing and subnetting
1.b	5	IPv4 addressing and subnetting
1.c	5	DHCP
1.d	5	NAT
1.e	5	Routing
2.a	8	Switch
2.b	5	ARP
2.c	5	VLAN
3.a	14	IPv4 Fragmentation
3.b	6	IPv4, ICMPv4
3.c	8	IPV6
Total	75	